

Borchester Urgent Care Centre Analysis Project

This project set out to find better ways to manage patient flow at Borchester Urgent Care Centre and improve the experience for both patients and staff. The aim was to cut down on waiting times, manage busy periods more smoothly, and make better use of the team's time and skills. Using real data from the clinic, a simulation was built to test a few different ways the system could work more efficiently.

The scenarios included keeping things as they are, bringing in an extra clinician, and introducing a triage step where patients are quickly assessed when they arrive. The results showed that even small adjustments can lead to meaningful improvements. While adding staff helped with delays, it wasn't always the most efficient choice when the clinic was quiet. The triage option stood out as a more balanced solution. It helped patients get to the right care faster and eased the pressure on the clinical team. While the model used some simplified assumptions, it still gave a realistic sense of what could work in everyday practice. In the end, the evidence pointed clearly toward triage as a practical, effective way to improve how the service runs.

Introduction

The Borchester Urgent Care Centre is essential to the local community, offering care for non-life-threatening issues like childhood fevers, sprains, and minor injuries. Like many walk-in clinics, it constantly juggles the challenge of matching staff availability with unpredictable patient numbers. On particularly busy days, the waiting area can quickly fill up, leading to long wait times, frustrated patients, and sometimes even people leaving before they've been seen. In contrast, quieter days can leave staff with little to do, which isn't the best use of resources. Diagram 1 shows how things are currently set up and identifies where the system gets overwhelmed.

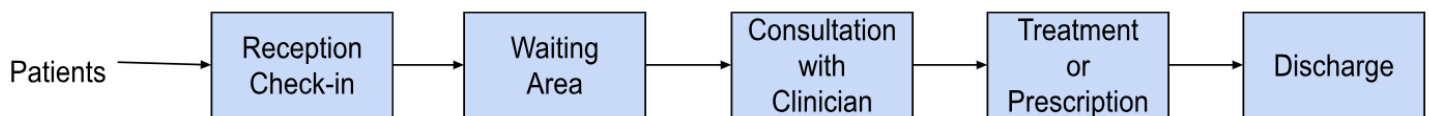
In response to these challenges, two potential strategies have been proposed. The first involves increasing clinical capacity by adding another healthcare provider. The second introduces a triage system, where a nurse conducts a quick initial assessment of each patient to determine whether they require simple or more complex care. Both approaches aim to improve patient flow and ease the strain on staff, particularly during peak periods.

This analysis explores how each strategy could impact key operational factors such as waiting times, queue lengths, and staff workload. Using real data from the clinic, we built simulations to test how these solutions might perform under varying conditions. Simulation modeling is a well established method in healthcare improvement work because it allows teams to explore the effects of operational changes in a risk-free environment. It offers a safe, cost-effective way to evaluate different approaches before committing resources to real world implementation.

This project is guided by a clear objective: to improve both patient care and staff experience. Inefficient management of patient flow in hospitals may result in negative health outcomes such as lower patient satisfaction, higher chances of readmission, and even increased mortality (Benjamin, 2024). Also, high

patient volumes and overcrowded conditions can significantly impact healthcare staff, contributing to elevated stress levels and a faster onset of burnout (Vile, 2017). These issues not only affect the well-being of staff but can also reduce the quality and consistency of care patients receive. While real world conditions can be unpredictable, this evidence based approach offers a strong foundation for designing a more balanced, responsive system. Improving flow within the clinic can help reduce unnecessary pressure, free up staff to focus on meaningful care, and make sure that patients receive timely attention in a setting that feels efficient, compassionate, and well managed.

Diagram 1. Current patient flow at Borchester Urgent Care Centre



Methods

To explore how patient flow at Borchester Urgent Care Centre could be improved, I used a two part approach: cleaning and preparing the data in Python and then using a queueing theory model in Excel. This combination made it possible to test different staffing and triage scenarios and see how each one might affect waiting times, queue lengths, and how clinicians are used.

I started by organizing the clinic's operational data in Python. This involved filtering out incomplete records, fixing time formats, and calculating how often patients arrive. Taking the time to clean the data properly helped make sure the model was built on accurate and reliable information. After preparing the data, I used Excel's built in queueing system tool to simulate patient movement through the clinic. The model follows the M/M/c queueing theory, which is commonly used in healthcare because it works well for systems like walk-in clinics. It assumes patients arrive randomly and that the time it takes to treat them varies, just like in real life. This made it a good fit for analyzing the flow of patients and staff in a busy, unpredictable setting.

Queueing theory was a strong choice for this project because it helped answer the main question: how can we make the clinic more efficient while still delivering high-quality care? The method made it easier to see how changes in capacity (like adding more staff), demand (patient arrival rate), and service delivery (how long treatment takes) could affect performance. By tweaking these inputs, I was able to explore the effects of various strategies under consistent and controlled conditions.

I built four different models:

Scenario 1: The current system, with two clinicians and no triage.

Scenario 2: One more clinician is added to the team.

Scenario 3 Clinic: A triage nurse sees all patients briefly before sending them to a clinician.

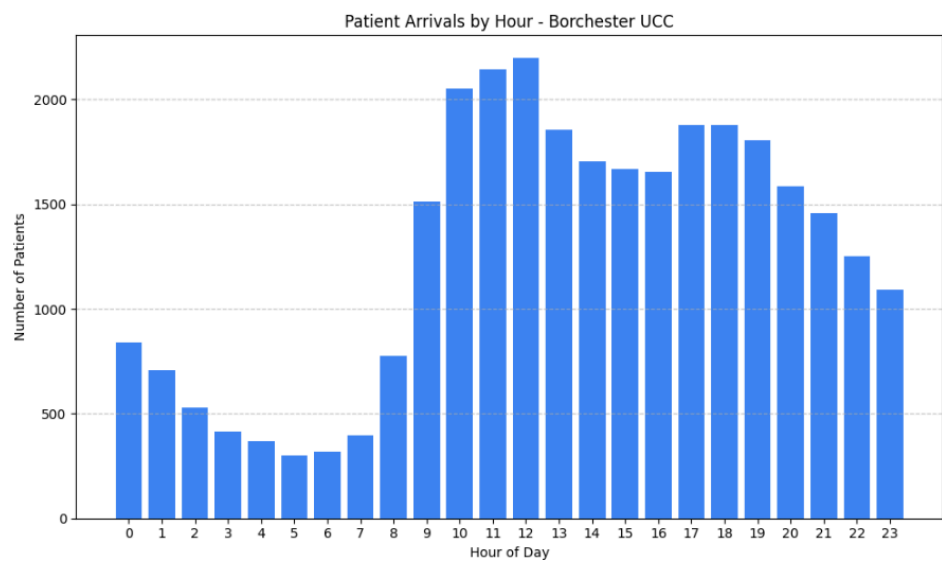
Scenario 3 Triage: Like Scenario 3, only 65% of patients are referred for treatment after triage.

The 65% figure in Scenario 3 was based on how triage tends to work in similar clinics. It's an estimate

that helped show how triage might ease pressure on clinicians if simpler cases can be handled at the front line. While the model simplifies things, it’s still a practical way to test the effects of different strategies without needing complex simulation software or more detailed patient level data. There are a few limitations to keep in mind. The model assumes that arrival and service rates stay constant and that every patient needs the same level of care. It also treats triage as perfectly accurate and consistent, which we know isn’t always the case. Still, these assumptions are common in operational planning and allow repeatable comparisons between options. This method gave a structured way to look at how patient flow and staffing could be improved. It provides managers with practical, data informed information to support decisions, especially during busy periods when the clinic is under the most strain.

Results

Graph 1: Patient Arrivals by Hour at Borchester UCC



Graph 1 shows the number of patient arrivals by hour. The data highlights that the busiest period tends to be from late morning to early afternoon, with arrivals peaking between 10 AM and 1 PM.

Table 1. Input Parameters and Output Measures for Baseline and Proposed Patient Flow Scenarios							
Scenario	Arrival Rate	Average Service Time (hours)	Capacity	Utilization	Queue Length (patients)	Waiting Time (hours)	Notes
Scenario 1	3.46	0.26	2	0.45	0.23	0.07	Current setup
Scenario 2	3.46	0.26	3	0.30	0.03	0.01	Added 1 clinician
Scenario 3 Clinician	3.46	0.08	1	0.29	0.12	0.03	All patients pass through triage
Scenario 3 Triage	2.25 (65% of 3.46)	0.26	2	0.29	0.05	0.02	Only 65% proceed to clinicians

Table 1 brings together the inputs and outputs for all four scenarios. It includes arrival rates, service times, clinician capacity, and the resulting queueing metrics. The baseline setup (scenario 1) which reflects the current clinic operations, includes two clinicians and no triage system. Under this model, the average waiting time was around 0.07 hours, with a queue length of 0.23 patients. Clinicians were working at about 45 percent capacity. This provided a useful benchmark to assess how changes might reduce delays or better distribute the workload.

Figure 1. Average Waiting Time by Scenario

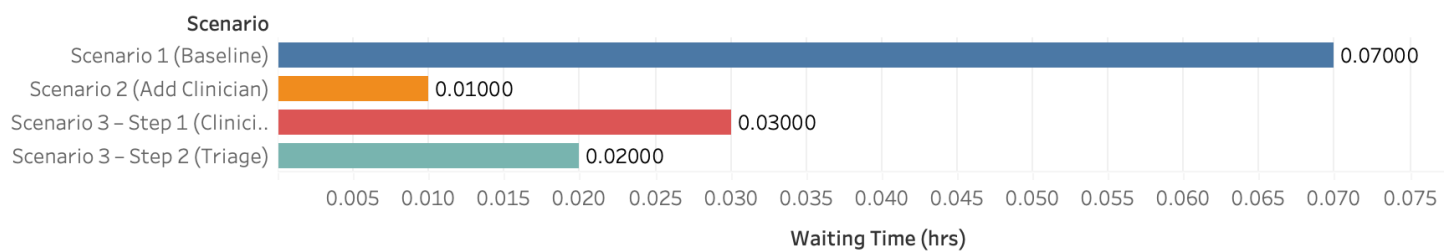


Figure 1 compares average waiting times for all four scenarios. Scenario 2, where one extra clinician is added, shows the shortest wait times. That’s expected, more staff naturally helps ease the load. But it’s not without a trade off. Adding more staff may reduce wait times, but it also means those clinicians might not be working at full capacity all the time.

Figure 2. Utilization Rate by Scenario

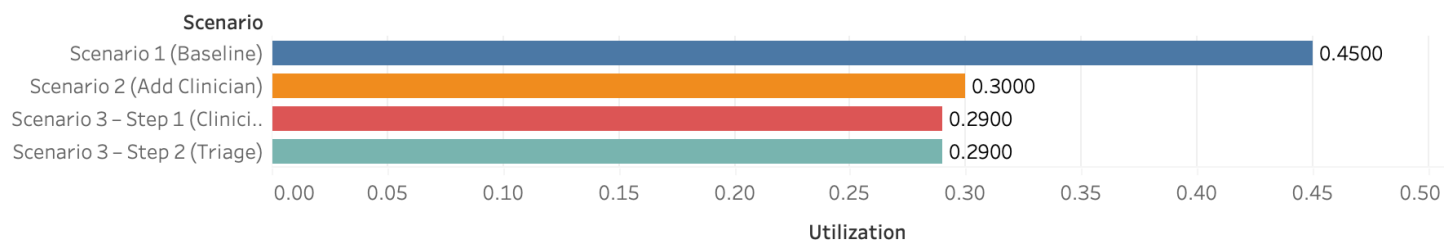
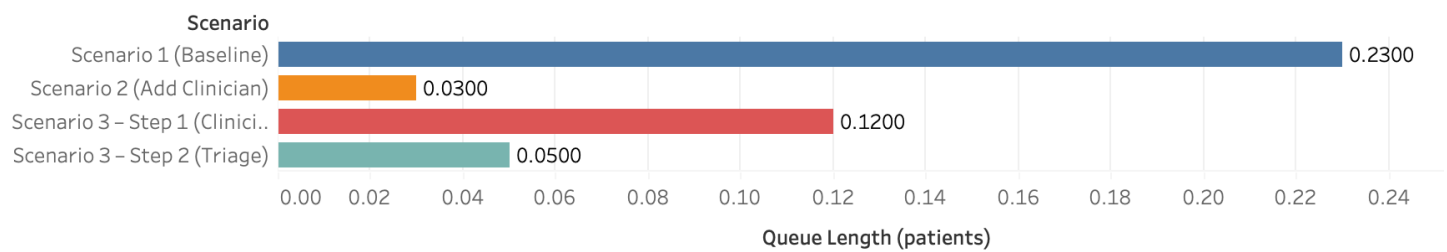


Figure 2 looks at clinician utilization. The baseline setup (S1) has the highest utilization, meaning staff are being used to their full capacity. While that sounds efficient, it also means there’s little room for unexpected surges in patient arrivals. Scenario 3, which includes a triage step where only around 65% of patients are referred on for full care, helps even things out. It keeps clinicians fairly busy, but also leaves a bit of flexibility in the system.

Figure 3. Queue Length by Scenario



In Figure 3, I compared queue lengths. The results line up with what we saw in the earlier metrics. The baseline setup has the longest queues, which can be frustrating for patients and stressful for staff. Scenario 2, again, performs best here, scenario 3 (Triage) also helps reduce queues, although it requires additional staff for triage duties.

Figure 4. Relationship Between Utilization and Waiting Time by Scenario



Figure 4 ties it all together by showing the relationship between staff utilization and patient waiting time. As expected, when clinicians are stretched thin, waiting times go up. Scenario 2 has the lowest waiting times, but also a drop in efficiency. On the other hand, Scenario 3 offers more balance, it brings down waiting times without requiring more staff.

Discussion

This analysis showed that small operational changes can make a meaningful difference in how the urgent care centre functions. The baseline setup had the highest staff workload and longest patient queues. Although this setup uses staff time efficiently during steady demand, it doesn't cope well with busy periods.

Adding a third clinician, as tested in Scenario 2, clearly improved service speed and queue lengths. However, during less busy times, staff were underutilized. This makes it less efficient overall and potentially more costly. While effective in reducing delays, this option may not be sustainable if resources are limited. Scenario 3 used a triage nurse to briefly assess patients before they see a clinician. This helped sort patients by need, keeping queues shorter and spreading the workload more evenly. Clinicians remained active but weren't overstretched. The trade off is that this approach still requires hiring one extra team member and assumes triage decisions are made accurately and consistently.

There are limitations to the model used in this project. It assumes patient arrivals and treatment times stay steady throughout the day and that all patients need the same level of care. It also simplifies the

triage process by assuming perfect accuracy, which may not reflect the reality of a busy clinic. These assumptions limit how precisely the model reflects day to day conditions, but they do allow for fair comparisons between options. Despite these constraints, the results point toward triage as the most balanced solution. It supports both patient flow and staff wellbeing, making it a practical and adaptable approach for improving urgent care delivery.

Recommendations

Data analysis strongly supports triage as a means of balancing patient wait times with clinician workload. The results from Scenario 3 show that having a triage nurse assess patients on arrival helps direct them to the right level of care more quickly, leading to shorter wait times and a more manageable workload for clinicians. This aligns with research by Subash et al. (2004), who found that team triage significantly cut down waiting times to see a doctor and get an X-ray in a Northern Ireland emergency department. That said, introducing triage does come with a few challenges. It means bringing in an additional staff member, but more importantly, it relies on quick, confident decisions to direct patients appropriately. If those decisions aren't consistent, there's a risk of misjudging who needs more urgent care. The good news is that these issues can be managed with the right training, clear guidelines, and regular check-ins to make sure the system is working as intended. With that support in place, triage has real potential to ease pressure on staff and help patients get seen more quickly and appropriately.

References

Excel was used to build and run a queuing model to simulate different scenarios within the clinic and assess the potential impact of proposed changes.

Python helped with data cleaning, processing, and initial exploratory analysis to better understand patient flow and staff utilization.

Tableau was used to create clear and engaging visualizations that highlight key patterns.

AI tools were used in a limited, assistive role, for proofreading and improving clarity. As a native English speaker, I relied on AI to polish the text rather than generate content.

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